Term paper 2

BAN403

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Inputs/Pictures/Logo NHH bl.png

1 Input analysis

1.1 Purpose

This section provides details for the stochastic inputs for the simulation in accordance with chapter 9 of the textbook (Laguna & Marklund, 2019). The purpose is to identify different elements of the Miller Pain Treatment Center (Chambers & Williams, 2017), organize the available data, fit appropriate probability distributions and justify all assumptions made in the process.

1.2 Inputs

The inputs from the Miller Pain Treatment Center can be categorized into three main groups; patient arrivals, treatment times and outcomes for patients (routing decisions). Each of these groups contains multiple stochastic variables that must be modelled probabilistically to capture the random nature of the system.

Patient arrivals are stochastic as they depend on patients arriving early, on time or late for their appointments. This results in queues and system congestion, which must be captured by the model.

Treatment times also vary between patients depending on their conditions and the complexity of the treatment needed. These variables include registration time, vitals, consultation time and treatment, as well as time spent on teaching residents.

Finally, patient outcomes provides stochastic routing decisions in the system. According to their needs, patients follow different paths through the system, resulting in varying resource utilization, waiting times and overall system performance.

All of these variables must be caught and modelled probailistically to ensure the simulation accurately depicts the system. Table 1 summarizes the stochastic variables identified in the Miller Pain Treatment Center, categorized into the three main groups:

Variable	Type	Description
Patient arrivals		
Arrival deviation	Continuous	Difference between actual arrival time and scheduled appointment time
Treatment times		
Registration time	Continuous	Time spent on administrative registration
Vitals time	Continuous	Time to record patient vitals
Physician assistant time	Continuous	Time spent by PA on follow-up patients
Check-out time	Continuous	Time to complete visit and paperwork
Resident review (new)	Continuous	Time reviewing patient file (new patients)
Resident–patient interaction (new)	Continuous	Consultation time with resident
Teaching time (new)	Continuous	Teaching interaction between resident and attending
Resident–attending interaction (new)	Continuous	Discussion between resident and attending
Resident review (return)	Continuous	Time reviewing file (return patients)
Resident–patient interaction (return)	Continuous	Consultation time with resident
Teaching time (return)	Continuous	Teaching interaction for return patients
Resident–attending interaction (return)	Continuous	Discussion between resident and attending
Patient outcomes (routing decisions)		
Patient type	Discrete	New, return, or follow-up patients
Follow-up requires attending	Discrete	Whether PA consults attending (approximately 50%)
No-show / cancellation	Discrete	Whether patient attends appointment (approximately 10% no-show)
Resident availability	Discrete	Number of residents available per session

Table 1: Identified stochastic inputs

References

Chambers, C., & Williams, K. (2017). Case—Miller Pain Treatment Center. *INFORMS Transactions on Education*, *17*(3), 121–127. https://doi.org/10.1287/ited.2017.0176cs

Laguna, M., & Marklund, J. (2019). *Business process modeling, simulation and design* (3rd). CRC Press.